
Toscanini 01 executive technology brief

invr452-fresh-evaporative-vaccine-coolerae7a00faa34a · run run · maturity EXPERIMENT_READY · 3D class ENGINEERING_3D

Technology

inv:r452-fresh-evaporative-vaccine-cooler:ae7a00faa34a

Bottleneck

- Problem Id: r452-fresh-evaporative-vaccine-cooler
- Device: passive evaporative-cooled vaccine storage vessel for last-mile clinics
- Failure: vaccine doses spoil in transit because the passive evaporative cooler loses cooling margin within hours when ambient humidity rises, the water-wick channel...

Causal mechanism

- Mechanism: A composite radiative coating embedded with microencapsulated phase-change materials that alter their infrared emissivity based on temperature thresholds
- Intervention: Apply a dual-layer coating where the base layer provides constant radiative cooling while the embedded phase-change l...

Differentiating architecture

- Collision Novelty Risk: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Prior Art Status: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Multi Source Directions: • Discovery: (sources_recorded, sources_errored)
- Destruction: (sources_recorded, sources_errored)
- Transfer: (sources_recorded, sources_errored)
- Reality: (s...

Expected technical advantage

UNKNOWN (not recorded)

Strongest supporting evidence

Evidence classification counts from the run's own verification: direct support: 3; partial support: 5; background: 1; analogy: 4; contradictory: 1; irrelevant: 1; direct-support spans are itemized in 04_EVIDENCE_SUMMARY.

Biggest uncertainty

FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005

Engineering status

EXPERIMENT_READY; 21 design inputs, 6 failure modes, 3 governing equations recorded

Decisive experiment

physical bench/clinical experiment — NOT simulation

Buyer decision

evaluate this package against the nine questions in 03_BUYER_DECISION_CARD; the cheapest falsification is named there with its acceptance rule

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.